Chao Tang

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Education

Tongji University Sep 2025 – Jun 2029 (Expected)

B.Eng. in Automation, Excellence Program

- **GPA: 4.52 / 5.00**; CET-4: 645; CET-6: 613.
- **Research Interests:** VLA, Deep Reinforcement Learning, Humanoid / Whole-Body Control, Adaptive Learning and Generalization.

Research & Projects

nSGS: Neural Success-Guided Sampling [GitHub] Oct 2026 – Present

- Developed **nSGS**, a state-conditioned neural task sampler for whole-body loco-manipulation on a **Go2 quadruped + Piper 6-DoF arm** (12 + 6 = 18-D action space). Trained PPO policies in Isaac Lab, adapting training command distributions based on student learning states and task feedback.
- Meta-trained the neural Teacher using **eight matched-start checkpoints**. Preliminary experiments across **four Student seeds** show a modest average improvement over Uniform sampling. Currently exploring cross-seed generalization, learning dynamics, and degradation-aware adaptive sampling.

$\pi_{0.5}$ **Fine-Tuning and RealMan Robot Manipulation** [GitHub] Nov 2026

- Fine-tuned $\pi_{0.5}$ on **50 teleoperated demonstrations** and completed an end-to-end robot learning pipeline, including data processing, policy training, inference, and deployment on a real RealMan robotic platform for a banana pick-and-place task.
- Compared fine-tuning strategies: **action-expert-only** fine-tuning achieved **9/10 successful onsite pilot trials**, while **LoRA** fine-tuning achieved 0/8.

Astra × RealMan: Multimodal Bimanual Manipulation [GitHub] Sep – Oct 2026

- Integrated **gpt6-astra** with RealMan dual-arm robots through multi-camera observations, a target-pose action interface, and execution feedback for closed-loop manipulation.
- Ran **four real-robot trials** ($\sim$30 min each), achieving **one complete pick–handoff–place** execution, and analyzed failure modes in spatial grounding and action execution.

ROS 2 Visual Manipulator [GitHub] May – Aug 2026

- Developed a ROS 2-based six-axis robot manipulation system, including joint-state feedback, serial control, gripper control, URDF/TF modeling, and a Qt interface.
- Integrated RGB-D perception, AprilTag, Open3D, inverse kinematics, and MoveIt 2 for object localization and grasp planning; performed integration and testing on physical hardware.

Industry Experience

AgiBot (Zhiyuan Robotics) Jul – Aug 2026

Mechanical Design Intern

- Participated in assembly, debugging, and continuous testing of G2 Air lift-mechanism fixtures; developed Python/pySerial scripts for automated servo cycling, state monitoring, and fault detection.
- Assisted with robot mechanical design and assembly verification, including structural documentation, bills of materials, and exploded-view drawings.

Technical Skills

- **Robot Learning:** Python, PyTorch, Deep Reinforcement Learning, PPO, Actor–Critic, Imitation Learning, Diffusion Policy, VLA Fine-Tuning.
- **Robotics & Simulation:** Isaac Lab, MuJoCo, LeRobot, ROS 2, MoveIt 2, URDF/TF, RGB-D Perception, Open3D.
- **Development & Hardware:** Linux, Git, C++ (basic), SolidWorks, Mechanical Design, Real-Robot Integration.